

INDIA-MADE DEFENCE LOGISTICS VTOL



Modular • Autonomous • Rugged • Mission-Ready

A proposed indigenous logistics platform inspired by the operational concept of long-range VTOL delivery systems, adapted for Indian defence logistics and surveillance support.

Presented by

SHIV-NETRA47 AIR DEFENCE SYSTEM PRIVATE LIMITED

Technology / Engineering Support: BCN Drone Center, Spain

MAKE IN INDIA

Defence Logistics • ISR Support • Resupply

TECHNICAL SUPPORT &
KNOWLEDGE TRANSFER BY

DRONE MANUFACTURING FROM **SPAIN** TO **INDIA**

INNOVATION. COLLABORATION. SELF-RELIANCE.

Building the Future of Aerial Technology Together



SPAIN APPROVED DRONE
European Standards | Certified Quality



MADE IN INDIA
Manufactured in India



**TECHNICAL SUPPORT &
KNOWLEDGE TRANSFER BY BCN**
Technology. Training. Transfer.

ENDURANCE VTOL | NANO DRONE

High Endurance. Long Range. Reliable.

-  **ENDURANCE** Up to 180 Min
-  **RADIO NETWORK** Up to 150 Km
-  **PAYLOAD CAPACITY** Up to 2.5 Kg
-  **ALL-WEATHER OPERATION**
Reliable in Tough Conditions
-  **APPLICATIONS**
Medicine | Logistics | Surveillance | Mapping



**FROM SPAIN
TO INDIA**



**BUILDING A STRONGER,
SELF-RELIANT INDIA**



TECHNICAL SUPPORT & KNOWLEDGE TRANSFER BY BCN
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BCN Drone Operation Medical Drone LOGISTICS Delivery



From Vein to Sky: Medical Drone Delivery

March 4, 2026

BCN Drone Operation

Medical Drone LOGISTICS Delivery Case Study



Despite significant technological progress over the past decade, routine medical drone deliveries remain rare in societies governed by stringent air-safety regulations, including Europe. The European Union Aviation Safety Agency (EASA) introduced amendments in September 2025 to the Acceptable Means of Compliance (AMC) and Guidance Material (GM) under Implementing Regulation (EU) 2019/947, which incorporate the latest version of the Specific Operations Risk Assessment (SORA) 2.5 framework developed by JARUS. This regulatory evolution has opened the door in some countries to more complex operational scenarios. Many view this as a precursor to fully urban drone deliveries under SAIL III. Under certain assumptions, such as the use of proven technology and duly authorized operators, this framework allows beyond visual line of sight (BVLOS) operations over population densities of up to 5,000 inhabitants per square kilometer. While restrictions on aircraft size still limit payload and range, routine operations delivering parcels of up to 5 kg within a 50 km radius are becoming feasible. This creates a compelling opportunity for regular drone corridors that connect regional hospitals with remote points of care, provided population density thresholds are met.

BCN Drone Operation Medical Drone LOGISTICS Delivery

SAIL III
Operational Authorisation
in Spain

Enabling advanced medical drone delivery with the **RigiTech Eiger**

SAIL III APPROVED
EASA SORA 2.5 COMPLIANT

SAFE BVLOS OPERATIONS
ADVANCED CONNECTIVITY
BUILT FOR HEALTHCARE IMPACT

REAL-WORLD IMPACT
Reliable. Efficient. Life-saving.

REGIONAL HOSPITAL **REGIONAL HOSPITAL**

CATUAV gains first Spanish SAIL III approval
July 1, 2026

BCN Drone CATUAV gains first Spanish SAIL III approval

We are proud to announce that CATUAV, the registered UAS operator branded as BCN Drone Center, has obtained the first SAIL III operational authorization granted in Spain.

This achievement culminates nearly a year of intensive work and close collaboration among key stakeholders across the UAS ecosystem:

AESA, as the competent authority, applied EASA's new SAIL III Means of Compliance alongside SORA 2.5 and guidance material developed jointly with industry.

RigiTech, as the designer and manufacturer of the Eiger 3 UAS, provided the technical basis required for the aircraft-specific approval.

CATUAV, as the UAS operator, led the operational implementation process.

The authorization, granted to CATUAV and linked to the RigiTech Eiger 3 platform, enables BVLOS drone delivery operations in urban environments. It represents an important step towards making intercity drone connections a practical and scalable reality.

One of the first operational applications is expected in southern Catalonia, where medical-delivery use cases are being developed with healthcare stakeholders to support more efficient, responsive and connected healthcare logistics networks.

This is not only a landmark achievement for CATUAV and BCN Drone Center, but also an important step in the continued maturity of the Spanish drone ecosystem. Beyond Spain, the authorization is among the first in Europe issued under SAIL III SORA 2.5 framework, demonstrating how higher-complexity BVLOS operations can progress from regulatory assessment to real-world deployment.



1. Strategic Opportunity

Why an indigenous VTOL logistics platform is relevant to Indian defence

FORWARD LOGISTICS

Deliver essential stores to remote, road-inaccessible or time-critical locations.

- Food & ration
- Medical supplies
- Spares & batteries
- Communications equipment
- Humanitarian / disaster relief

OPERATIONAL ADVANTAGE

Vertical take-off reduces dependence on prepared runways while fixed-wing cruise can extend efficiency and range.

- VTOL + cruise
- Autonomous route execution
- Precision landing
- Day / night capable architecture

INDIAN REQUIREMENT FIT

India's iDEX ecosystem has explicitly identified heavy-payload logistics drones, including a 200 kg swarm logistics challenge.

This proposal is positioned as an indigenous, modular response.

2. Reference Architecture — RigiTech Eiger

Benchmark only: the proposed Indian system is not a copy and requires its own engineering, trials and certification

REFERENCE: EIGER

Publicly stated Eiger characteristics:

- VTOL + fixed-wing
- 3 kg max payload
- 100 km max range
- BVLOS-oriented autonomy
- Precision landing
- Parachute / failsafe
- Cloud mission management

INDIA-MADE DIRECTION

Scale the architecture toward defence logistics:

- 50–100 kg initial prototype class
- 200 kg scalable target architecture
- Modular cargo bay
- Rugged landing / transport system
- Secure mission management
- Indian supply-chain localization

DESIGN PRINCIPLE

Separate the aircraft, mission computer, communications, cargo module and ground station into replaceable modules.

This enables a family of aircraft rather than a single fixed configuration.

All performance figures are proposed engineering targets until validated by flight testing.

3. Proposed India-Made Platform — Initial Specification

Conceptual target values for Phase-1 engineering; final specifications to be established through design and trials

Aircraft

VTOL + fixed-wing hybrid

Payload

5–10 kg Phase-1 target; 200+ kg scalable architecture

Mission radius

20–50 km initial target; longer-range variants subject to design 100+ possible

Flight mode

Manual / assisted / autonomous

Navigation

GNSS + INS; NAVIC-ready architecture

Landing

Precision landing on prepared or semi-prepared sites

Cargo

Quick-change enclosed modular payload bay

Safety

Independent failsafe logic; emergency recovery system to be evaluated

Environment

Ruggedized architecture for heat, dust, rain and field deployment

Ground system

Portable GCS + mission planning + fleet monitoring

4. Defence Logistics Mission Concept

A non-weaponized logistics and surveillance-support architecture



Optional mission payloads: EO/IR situational-awareness camera, mapping sensor, communications relay, medical module or other approved non-weaponized payloads.

5. System Architecture

Modular design for Indian manufacturing, maintenance and future upgrades

AIRFRAME

Composite / hybrid structure
VTOL lift + cruise wing

FLIGHT CONTROL

Autopilot • navigation • health monitoring

MISSION COMPUTER

Route management • autonomy • payload interface

COMMUNICATIONS

Redundant command / telemetry architecture

CARGO MODULE

Quick-change logistics container / mission bay

GROUND SYSTEM

Portable GCS • mission planning • fleet dashboard

6. Proposed India–Spain Collaboration Model

SHIV-NETRA47 as Indian prime / system integrator with BCN technical and UAS ecosystem support

SHIV-NETRA47 INDIA



- Programme leadership
- Indian defence interface
- System integration
- Local manufacturing
- Supplier development
- Field trials in India
- Lifecycle support

BCN DRONE CENTER SPAIN



- UAS engineering support
- Prototype development support
- Test methodology
- Training & pilot capability
- Regulatory / operational know-how
- Access to European UAS ecosystem
- International demonstration support

JOINT OUTCOME

- India-made VTOL platform
- Transfer of know-how
- Local assembly → localization
- Test & evaluation programme
- Defence / civil logistics variants
- Export-ready architecture subject to approvals

7. Make in India — Development & Localization Roadmap

Proposed phased programme

PHASE 1

0–1 MONTHS

Mission definition
Architecture
Digital design
Subsystem selection
Bench testing

PHASE 2

1–3 MONTHS

Prototype build
Ground tests
Flight-control integration
Initial flight trials

PHASE 3

3–6 MONTHS

Payload trials
Reliability testing
Environmental trials
Operational demonstrations

PHASE 4

6–12 MONTHS

Pre-production
Supplier localization
Training & maintenance
User acceptance trials

Localization target: establish an Indian supply chain for structure, wiring, avionics integration, ground equipment and maintenance; exact percentages to be fixed after BOM and technology audit.

8. Indian Production & Sustainment Model

Designed for scalable assembly and field maintenance

MANUFACTURING

- Composite / structural work
- Mechanical assembly
- Electrical harness
- Avionics integration
- Ground-station build
- QA / traceability

SUPPLY CHAIN

- Indian electronics & machining
- Motors / ESC / power systems
- Batteries & charging
- Sensors & communications
- Standardized fasteners / spares
- Dual-source critical items

FIELD SUPPORT

- Modular line-replaceable units
- Quick diagnostics
- Battery health monitoring
- Operator training
- Maintenance manuals

SCALE

Target family:

- Logistics VTOL
- Heavy-lift variant
- ISR-support variant
- Disaster-relief variant
- Swarm-compatible architecture

Subject to approvals & testing.

9. Why This Program Fits the Current Indian Defence Innovation Landscape

- iDEX has active defence innovation programmes focused on unmanned systems and logistics.
- India's defence innovation framework is designed to bring startups, MSMEs and innovators into indigenous defence technology development.
- BCN Drone Center provides UAS development, testing, training and operational ecosystem support in Europe.
- SHIV-NETRA47 can position the program as an India-led, modular logistics platform with a clear path from prototype to production.

10. Proposed Demonstration Programme

A controlled, measurable route from concept to user evaluation

A	Prototype	1–2 aircraft in initial payload class
B	Ground Trial	Power, navigation, cargo, communications, failsafe
C	Flight Trial	VTOL, transition, cruise, landing, return-to-base
D	Logistics Demo	Repeatable point-to-point supply mission
E	Evaluation	Reliability, maintainability, range, payload and turnaround

Success criterion: demonstrate a safe, repeatable and maintainable logistics workflow—not merely a single flight.

FROM CONCEPT TO INDIA-MADE CAPABILITY

SHIV-NETRA47 × BCN DRONE CENTER

Develop • Test • Localize • Demonstrate • Scale

INDIA

System integration
Manufacturing
Defence interface

EUROPE

UAS know-how
Testing support
Training

OUTCOME

Indigenous logistics
platform family

Prepared as a concept proposal — specifications and capabilities require engineering validation, trials, certification and customer requirements.